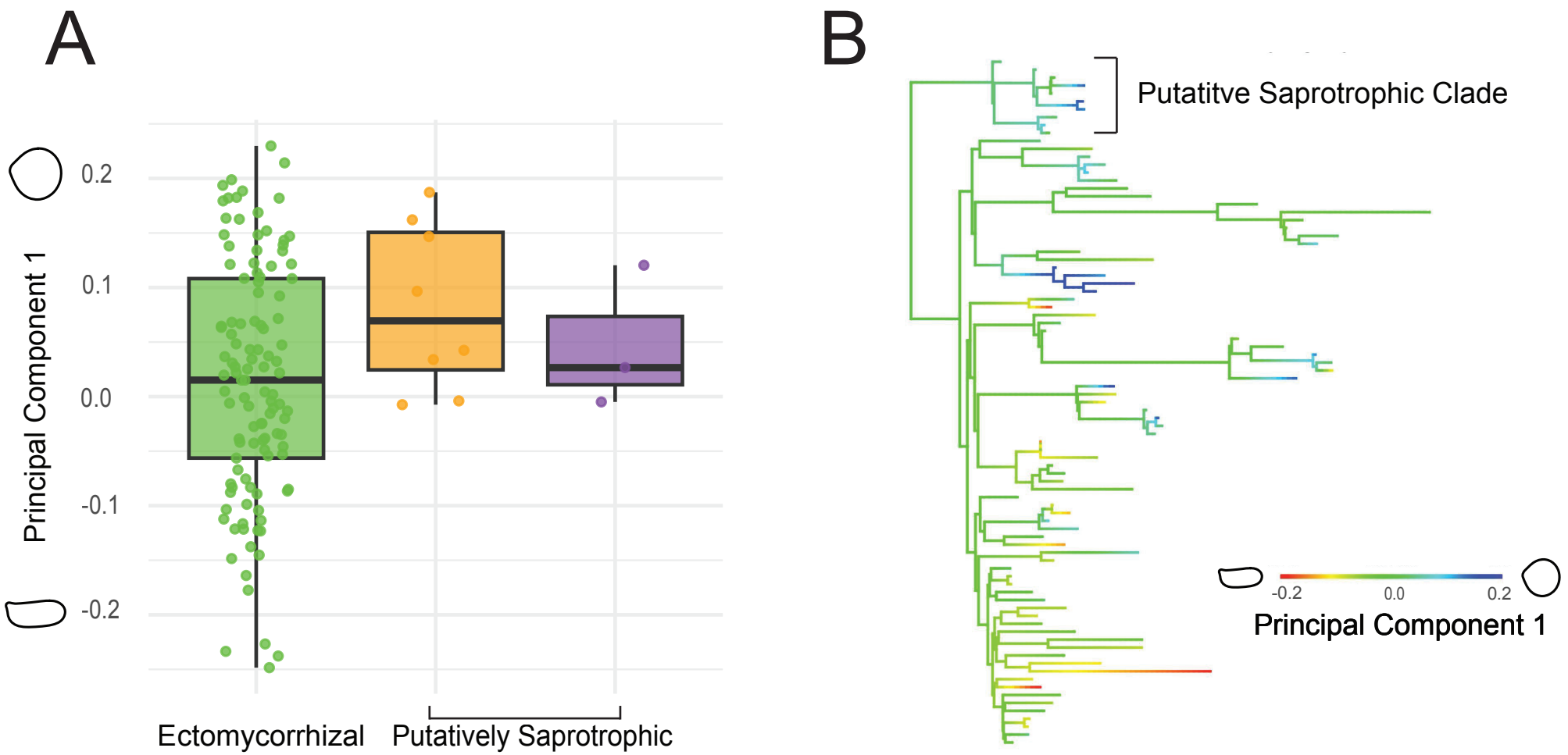


Figure. 4 Analysis of the influence of trophic mode on principal component 1 (PC1). Trophic mode refers to the distinction between ectomycorrhizal (EM) *Amanita* that are obligate symbionts, and a clade of putatively saprotrophic (SAP) *Amanita*, consisting of the clade Inopinata (a member of Amanitina) and subgenus Lepidella. (a) Naive comparison of trophic mode and PC1. Clade Inopinata is shown in purple, subgenus Lepidella is shown in orange, and the rest of the genus *Amanita* is in green. The difference between these two groups is weakly non-significant ($p = 0.08$) when using designations based on genetic phylogeny. (b) Phylogenetic tree of *Amanita* overlaid with PC1. When accounting directly for phylogenetic relatedness, there is no difference between EM and SAP groups (phyloANOVA: $P = 0.449$) in regards to PC1. Moreover, over the entire genus *Amanita*, there is no phylogenetic signal and the PC1 trait appears to evolve unpredictably and not gradually (Blomberg's K: 0.000002, Pagel's λ: 0).